

AtIPL: Sheet Four

① Big Q. So best make sure we unpack it right to ensure we answer it

constants and functions fragment

excellent this will save us time as we only need to deal with

Static

Dynamic

LAM
APP

D-BETA
D-APP-1
D-APP-2

(VAL-LAM) but not when introducing $\rightarrow$

(as the hint says, last week's proofs did the rest of the fragment)

The Q even suggests us steps!

Plan

1. extend

- injection

- subst

- canonical forms

to function type

2. Assume weakening

3. Prove progress

4. Prove preservation

CLAIM 1

CLAIM 2

CLAIM 3

ASS-WK

CLAIM 4

CLAIM 5

CLAIM 1. (Inversion):

Suppose $\Gamma \vdash e : \tau$
 (LAM) If $e = \lambda x : \sigma. e'$ then $\tau = \sigma \rightarrow \tau'$ for
 some type τ' , and $\Gamma, x : \sigma \vdash e' : \tau'$
 (APP) If $e = e_1 (e_2)$ then there exists a τ_2
 s.t. $\Gamma \vdash e_1 : \tau_2 \rightarrow \tau_1$ and $\Gamma \vdash e_2 : \tau_2$

this is precisely what the rules say.

Proof by inspection

$$\text{LAM} \frac{\Gamma \vdash x : \sigma \vdash e : \tau}{\Gamma \vdash \lambda x : \sigma. e : \sigma \rightarrow \tau}$$

To have the judgement $\Gamma \vdash \lambda x : \sigma. e : \sigma \rightarrow \tau$, we must have a derivation for $\Gamma \vdash x : \sigma \vdash e : \tau$.

In other words LAM-INV

$$\text{APP} \frac{\Gamma \vdash e_1 : \sigma \rightarrow \tau \quad \Gamma \vdash e_2 : \sigma}{\Gamma \vdash e_1 (e_2) : \tau}$$

To have a judgement $\Gamma \vdash e_1 (e_2) : \tau$, we must have derivations for $\Gamma \vdash e_1 : \sigma \rightarrow \tau$ and $\Gamma \vdash e_2 : \sigma$

In other words APP-INV.

CLAIM 1
(Inversion)

CLAIM 2 (substitution):

If $\Gamma \vdash e : \tau$ and $\Gamma, x:\tau \vdash u : \sigma$
 then $\Gamma \vdash u[e/x] : \sigma$

We have already shown this for the previous parts of the lab (lecture 4 notes, sheet 2) so we know this is the proof by induction on our ~~proof~~ ~~proof~~.

If we didn't already know we could look and see we have two premises of the same form and pick the more exciting one.

Proof by induction on $\Gamma, x:\tau \vdash u : \sigma$

(remember only doing function stalk rules of LAM and APP)

[Case: LAM]

$\Rightarrow \Gamma, x:\tau \vdash u : \sigma$ has shape $\Gamma, x:\tau \vdash \lambda y. \sigma_1. u' : \sigma_1 \rightarrow \sigma_2$
 and derivation

: SHAPE

$$\frac{\Gamma, x:\tau, y:\sigma_1 \vdash u' : \sigma_2}{\Gamma, x:\tau \vdash \lambda y. \sigma_1. u' : \sigma_1 \rightarrow \sigma_2} \text{LAM}$$

smaller same shape $\Rightarrow$ IH u: σ specialised with one less chosen var names towards conflict

IH = if $\Gamma, y:\sigma_1 \vdash e : \tau$ and $\Gamma, x:\tau, y:\sigma_1 \vdash u' : \sigma_2$
 then $\Gamma, y:\sigma_1 \vdash u'[e/x] : \sigma_2$

ASS (other premise) = $\Gamma \vdash e : \tau$

GOAL = $\Gamma \vdash u[e/x] : \sigma$
 $= \Gamma \vdash (\lambda y. \sigma_1. u')[e/x] : \sigma_1 \rightarrow \sigma_2$

: ASS

WK $\frac{\Gamma \vdash e : \tau}{\Gamma, y:\sigma_1 \vdash e : \tau}$: SHAPE

$$\frac{\Gamma, y:\sigma_1 \vdash e : \tau \quad \Gamma, x:\tau, y:\sigma_1 \vdash u' : \sigma_2}{\Gamma, y:\sigma_1 \vdash u'[e/x] : \sigma_2} \text{IH}$$

$$\frac{\Gamma, y:\sigma_1 \vdash u'[e/x] : \sigma_2}{\Gamma \vdash (\lambda y. \sigma_1. u'[e/x]) : \sigma_1 \rightarrow \sigma_2} \text{LAM}$$

$$\frac{\Gamma \vdash (\lambda y. \sigma_1. u'[e/x]) : \sigma_1 \rightarrow \sigma_2}{\Gamma \vdash (\lambda y. \sigma_1. u') [e/x] : \sigma_1 \rightarrow \sigma_2} \text{def subst}$$

□ LAM

[Case: APP]

$\Rightarrow \Gamma \vdash u : \sigma$ has shape $\Gamma \vdash u_1 (u_2) : \sigma$
 and derivation

: SHAPE 1 : SHAPE 2

$$\frac{\Gamma, x:\tau \vdash u_1 : \tau_1 \rightarrow \sigma \quad \Gamma, x:\tau \vdash u_2 : \tau_1}{\Gamma, x:\tau \vdash u_1 (u_2) : \sigma} \text{APP}$$

IH1 = if $\Gamma \vdash e : \tau$ and $\Gamma, x:\tau \vdash u_1 : \tau_1 \rightarrow \sigma$
 then $\Gamma \vdash u_1 [e/x] : \tau_1 \rightarrow \sigma$

IH2 = if $\Gamma \vdash e : \tau$ and $\Gamma, x:\tau \vdash u_2 : \tau_1$
 then $\Gamma \vdash u_2 [e/x] : \tau_1$

ASS (other premise) = $\Gamma \vdash e : \tau$

GOAL = $\Gamma \vdash u[e/x] : \sigma$
 $\Gamma \vdash u_1 (u_2) [e/x] : \sigma$

: ASS : SHAPE 1 : ASS : SHAPE 2

$$\frac{\Gamma \vdash e : \tau \quad \Gamma, x:\tau \vdash u_1 : \tau_1 \rightarrow \sigma \quad \Gamma \vdash e : \tau \quad \Gamma, x:\tau \vdash u_2 : \tau_1}{\Gamma \vdash u_1 [e/x] (u_2 [e/x]) : \sigma} \text{IH1, IH2}$$

$$\frac{\Gamma \vdash u_1 [e/x] (u_2 [e/x]) : \sigma}{\Gamma \vdash u_1 (u_2) [e/x] : \sigma} \text{subst}$$

□ APP

□ CLAIM 2 (subst)

CLAIM 3 (canonical forms):

If $\vdash e : \sigma \rightarrow \tau$ and $e \text{ val}$

then $e = \lambda x : \sigma. u$ with $x : \sigma \vdash u : \tau$

this is another just about the message

Proof by inspection.

VAL-LAM

$\lambda x : \tau. e \text{ val}$

LAM

$\frac{\mathbb{P}, x : \sigma \vdash e : \tau}{\mathbb{P} \vdash \lambda x : \sigma. e : \sigma \rightarrow \tau}$

$\mathbb{P} \vdash \lambda x : \sigma. e : \sigma \rightarrow \tau$

A function is a value
this gives us the first
part of our conclusion (1)

In order to conclude
that $\mathbb{P} \vdash \lambda x : \sigma. e : \sigma \rightarrow \tau$
we must have had
 $\mathbb{P}, x : \sigma \vdash e : \tau$. this gives
us (2).

CLAIM 4 (Progress):

If $t e : \tau$ then either eval or $e \mapsto e'$ for some e'

Having done this before, we again know this to be a proof by induction on our antecedent $t e : \tau$

Proof by induction on $t e : \tau$

[Case: LAM]

$\Rightarrow t e : \tau$ has shape $\vdash \lambda x : \sigma. e : \sigma \rightarrow \tau$ and derivation

$$\frac{\begin{array}{c} \vdash \text{SHAPE} \\ x : \sigma \vdash e : \tau \end{array}}{\vdash \lambda x : \sigma. e : \sigma \rightarrow \tau} \text{LAM}$$

IH = if $x : \sigma \vdash e : \tau$ then either eval or $e \mapsto e'$ for some e'

GOAL = either $\lambda x : \sigma. e$ eval or $\exists e' \text{ s.t. } \lambda x : \sigma. e \rightarrow e'$

The goal can actually just be proved with VAL-LAM, which has no premises. Still good to write out the IH and shape in case we needed them

$$\frac{}{\lambda x : \sigma. e \text{ eval}} \text{VAL-LAM}$$

$\square \text{LAM}$

[Case: APP]

$\Rightarrow t e : \tau$ has shape $\vdash e_1 (e_2) : \tau$ and derivation

$$\frac{\begin{array}{c} \vdash \text{SHAPE 1} \\ \vdash e_1 : \tau \rightarrow \sigma \end{array} \quad \begin{array}{c} \vdash \text{SHAPE 2} \\ \vdash e_2 : \tau \end{array}}{\vdash e_1 (e_2) : \sigma} \text{APP}$$

IH 1 = if $\vdash e_1 : \tau \rightarrow \sigma$ then either e_1 eval or $\exists e_1' \text{ s.t. } e_1 \mapsto e_1'$

IH 2 = if $\vdash e_2 : \tau$ then either e_2 eval or $\exists e_2' \text{ s.t. } e_2 \mapsto e_2'$

GOAL = either $e_1 (e_2)$ eval or $\exists e' \text{ s.t. } e_1 (e_2) \rightarrow e'$

Thinking there is no rule to eval this immediately, so we must use our IHs. To use these we must case split on the different possibilities of their result, proving our goal in each case.

[Subcase: e_1 eval]

$e_1 : \tau \rightarrow \sigma$, so by canonical forms, it must be

$e_1 = \lambda x : \tau. u$ for some u .

So we have $(\lambda x : \tau. u) e_2$ which we either need to show to be a val, or produce a reduct for.

Applications are not values as there will still work to do, so we will show the latter.

Happily the beta rule with no premises matches our case.

$$\frac{\text{D-BETA}}{(\lambda x : \tau. u) e_2 \mapsto \underbrace{u[e_2 / x]}_{e'}} \quad \square \text{eval}$$

[Subcase: $\exists e_1 \text{ s.t. } e_1 \mapsto e_1'$]

Again this ain't a value so we look for an appropriate $\mapsto$ rule to find a reduct of $e_1 (e_2)$.

$\square \text{APP}$

$$\frac{\text{D-APP-1} \quad e_1 \mapsto e_1'}{e_1 (e_2) \mapsto e_1' (e_2)}$$

Using just IH 2 in either case we get our goal $\Rightarrow \square$ $\square \exists e' \text{ s.t. } e \rightarrow e'$

CLAIM 5 (Preservation):

If $\vdash e : \tau$ and $e \mapsto e'$
then $\vdash e' : \tau$

Hopefully you know the drill by now. We induct on $e \mapsto e'$ (that is chosen as it mentions e' making it the most relevant premise)

Proof by induction on $e \mapsto e'$

this will be 3 cases: D-APP-1, D-APP-2 and D-BETA. (VAL-LAM is not a $\mapsto$ judgement)

[Case: D-APP-1]

$\Rightarrow e \mapsto e'$ has shape $e_1(e_2) \mapsto e'_1(e_2)$

and derivation

$$\frac{\vdash \text{SHAPE} \quad e_1 \mapsto e'_1}{e_1(e_2) \mapsto e'_1(e_2)} \text{D-APP-1}$$

IH = if $\vdash e_1 : \sigma$ and $e_1 \mapsto e'_1$
then $\vdash e'_1 : \sigma$

GOAL = $\vdash e'_1(e_2) : \tau$

ASS = $\vdash e_1(e_2) : \tau$

By inversion on ASS, we know

$e_1 : \tau_2 \rightarrow \tau$ (INV1)
 $e_2 : \tau_2$ (INV2)

$\vdash \text{INV1} \quad \vdash \text{SHAPE} \quad e_1 \mapsto e'_1$

$\vdash \text{INV2} \quad e'_1 : \tau_2 \rightarrow \tau \quad e_2 : \tau$

$\vdash e'_1(e_2) : \tau$

Alternatively you can start deriving goal, see what is needed and try find your derivation ingredients. I just like the recipe nature of case into shape ASSs/IHs Goal.

I really am in the habit of just unceremoniously cutting all helpful things before trying to derive

and having done that we derive the goal just focus on

APP $\square$ D-APP-1

[Case: D-APP-2]

Analagous to D-APP-1.

[Case: D-BETA]

$\Rightarrow e \mapsto e'$ has shape $(\lambda x : \tau. e_1) e_2 \mapsto e_1[e_2/x]$

and derivation

$$\frac{}{(\lambda x : \tau. e_1) e_2 \mapsto e_1[e_2/x]} \text{D-BETA}$$

ASS(antecedent) = $\vdash (\lambda x : \tau. e_1) e_2 : \sigma$

GOAL = $\vdash e_1[e_2/x] : \sigma$

This looks very like the substitution lemma

By the substitution lemma to conclude

$\vdash e_1[e_2/x] : \sigma$

(need:

- $\vdash e_2 : \tau$ (OBLIGATION 1)
- $x : \tau \vdash e_1 : \sigma$ (OBLIGATION 2)

These can both be discharged by inversion on ASS

$$\text{LAM} \frac{x : \tau \vdash e_1 : \sigma}{\vdash (\lambda x : \tau. e_1) : \tau \rightarrow \sigma} \quad \frac{\vdash (\lambda x : \tau. e_1) : \tau \rightarrow \sigma \quad \vdash e_2 : \tau}{\vdash (\lambda x : \tau. e_1) e_2 : \sigma} \text{APP}$$

$\square$ D-BETA

$\square$ CLAIM 5 (preservation)

② "Do we have to prove progress/preservation?"
is certainly the question on everyone's lips,
but maybe not for the right reasons. I'd

Well, let's think about what has changed:
the dynamics

$\Rightarrow$ if the proof relies on dynamics we
need to make some adjustment.

Preservation

If $t \vdash \tau$ and $e \mapsto e'$ then $t \vdash \tau$

Now, this is a proof by induction on the
dynamics ($\mapsto$). $\Rightarrow$ the proof would need
reworking with the new rules and adjustments
to the rule. Having said that, we can
certainly use the old proof to our
advantage.

It must be proved for the new cases:

D-INTL
D-INTR
D-APP-2

The changed rules also need to be checked to
see if the adjustments check the proof.

Progress

If $t \vdash \tau$ then either eval or $e \mapsto e'$ for some e'

this is a proof by induction on the statics,
so the statics won't change, but we
need to conclude either a value or a
transition, so our conclusions may need
adjusted.

Some transition conclusions may need
adjusted.